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ODD END SEMESTER EXAMINATION DECEMBER – 2024

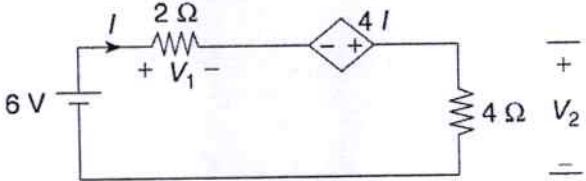
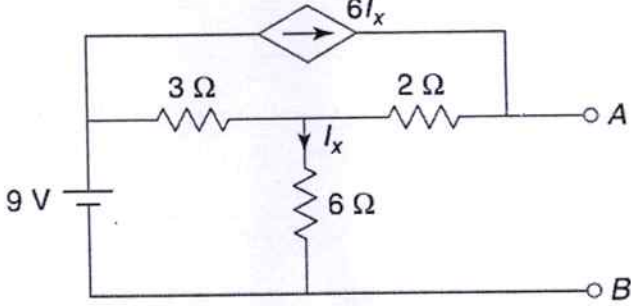
Name of the Course: B.TechSemester: 3rdName of the Paper: Electrical Circuit AnalysisPaper Code: TEE 301

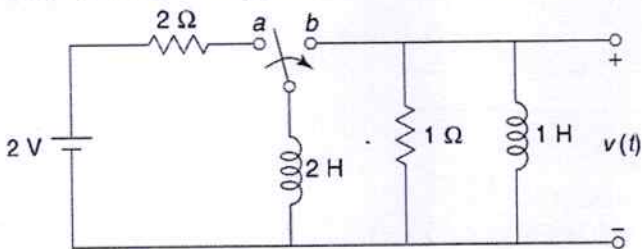
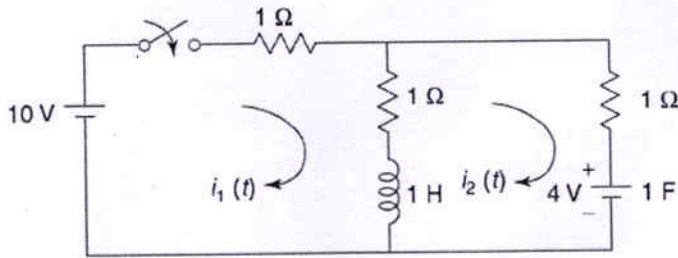
Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Define the following terms. (i) Thevenin's Theorem (ii) Voltage controlled current source (iii) Voltage controlled voltage source (iv) Independent voltage source (v) Independent Current source	CO2 CO3
(b)	Find the voltage v_1 and v_2 in the Fig. 1  Fig. 1.	
(c)	Find V_{Th} and R_{Th} between terminals A and B of the network shown in Fig. 2  Fig. 2	
Q2	(20 marks)	CO2, CO4

(a)	An inductive coil having negligible resistance and 0.1 henry inductance is connected across a 200 V, 50 Hz supply. Find (a) inductive reactance, (b) RMS value of current, (c) power, (d) power factor, and (e) equations for voltage and current.	
(b)	Three coils, each having a resistance of 8 ohm and an inductance of 0.02 H, are connected in delta to a three-phase, 400 V, 50 Hz supply. Calculate the (a) line current, and (b) power absorbed.	
(c)	Derive the relationship between power in delta and star system.	
Q3	(20 marks)	
(a)	<i>Solve $y'' + y = t$, $y(0) = 1$, $y'(0) = 0$.</i>	CO5
(b)	The network of Fig. 3 was initially in the steady state with the switch in the position a. At $t=0$, the switch goes from a to b. Find an expression for voltage $v(t)$ for $t > 0$.  Fig. 3	
(c)	In the network shown in Fig. 4, the switch is closed at $t=0$. Find the currents $i_1(t)$ and $i_2(t)$ when initial current through the inductor is zero and initial voltage on the capacitor is 4V.  Fig. 4	
Q4	(20 marks)	CO 6
(a)	Explain the condition of Reciprocity and Symmetry in two port networks. Derive the relationship in Z-parameters for Reciprocity and Symmetry conditions.	
(b)	Draw the equivalent circuit of the two-port network in terms of (i) Z-parameters (ii) Y-parameters	

(c)

Find the Z-parameters for the network shown in Fig. 5

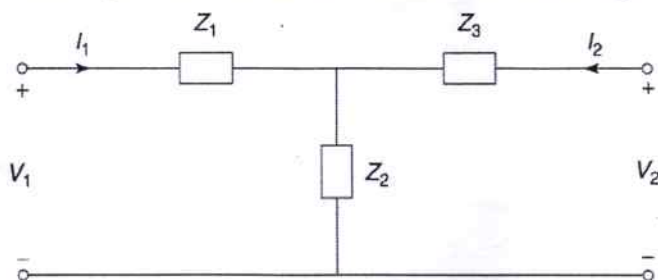


Fig. 5

Q.5

(20 marks)

(a)

Find the response voltage in the network shown in Fig. 6. Using Fourier Transform method.

CO5

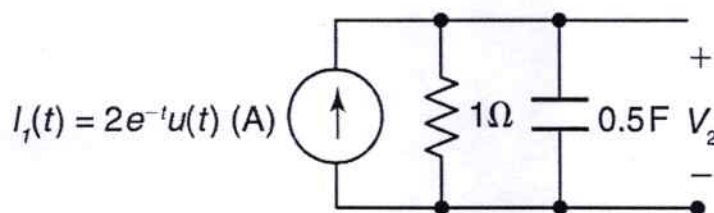


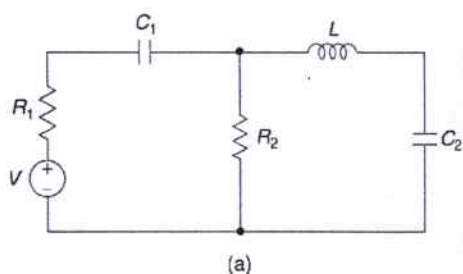
Fig. 6

(b)

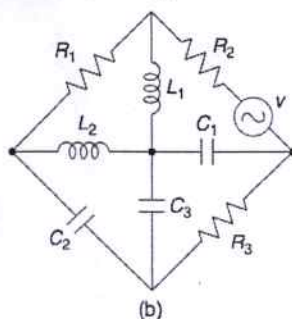
A series RL circuit with $R = 10 \Omega$ and $L = 5 H$ contains a current $i(t) = 10 \sin 1000 t + 5 \sin 3000 t + 3 \sin 5000 t$. Find the effective value of voltage and the average power.

(c)

Draw the directed graph of the network shown in Fig. 7



(a)



(b)

Fig. 7

